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Prevalence of Extended-Spectrum Beta-Lactamase Resistance and CTX-M-Group 1 Gene in *Escherichia coli* from the Water and Sediment of Urbanized Mangrove Ecosystems of Kerala

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The study aimed to determine the prevalence of extended-spectrum β -lactamase resistance and CTX-M-group 1 gene in *Escherichia coli* from the water and sediment of three urbanized mangrove ecosystems of Kerala. A total of 119 *E. coli* isolates were screened for antibiotic susceptibility to 16 antibiotics. According to the phylogenetic analysis of *E. coli* isolates, nonpathogenic group A and pathogenic group D (29.4% and 23.5%) were the predominant phylotypes found in water samples. The most frequent phylotypes found in sediment samples were nonpathogenic groups A and B1 (27.9% and 26.4%). The highest incidence of antibiotic resistance in *E. coli* was against cefotaxime and colistin (100%). A significant difference in the prevalence of CTX-M-group 1 gene was observed among *E. coli* isolates in water samples ($p < 0.05$). The results indicate a high prevalence of β -lactamase harboring *E. coli* in the mangrove ecosystems that can hamper mangrove-dependent aquaculture practices and human health.

Keywords: antibiotic resistance, phylogenetic groups, beta-lactam, mangrove

Introduction

INFECTIONS CAUSED BY MULTIDRUG-RESISTANT pathogens are a challenging global epidemic that reinforces the necessity of the “One Health” paradigm.¹ Inappropriate prescription and over consumption of antibiotics have accelerated antibiotic resistance globally and have outpaced treatment options.^{2–4} Owing to the discharge of untreated hospital effluents and inadequate sanitation technology, underdeveloped and emerging countries form the significant hotspots for antibiotic-resistant bacteria (ARB).^{5–7}

Continuous antibiotic prescription in human and veterinary health and the lack of effective regimes to control or prevent infections have led the way to the increased occurrence of β -lactamase enzymes.^{8,9} β -Lactamases are the bacterial enzymes that are considered the prime cause of bacterial resistance against drugs such as cephalosporin and carbapenems, and they protect the microbe against the lethal

actions of β -lactam antibiotics by disrupting the indispensable β -lactam bond.^{10,11} These enzymes can be transferred among and between bacterial species.^{12,13}

The production of extended-spectrum β -lactamase (ESBL) causes nonsusceptibility to various β -lactam agents. ESBL-producing Enterobacteriaceae are frequently coresistant to a range of different classes of antibiotics, further exacerbating the therapeutic confrontation.¹⁴ *Escherichia coli* is an ubiquitous member of the intestinal flora of humans and animals, can easily contaminate the surrounding environments when discharged along with feces, and is often used as an indicator organism to determine the prevalence of antibiotic resistance worldwide.^{15–17}

They are considered as an important contamination indicator in drinking water, aquatic systems, and foods.^{17–20} Antibiotic resistance in *E. coli* arises due to mutations, acquisition of mobile genetic elements or horizontal gene transfer, and can spread by cross-reservoir transmission through the food chain.²¹

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In recent years, many studies have revealed the presence of ARB in different environments including mangroves, aquaculture ponds, and platisphere.^{22–28} Among them, mangroves are highly productive “natural wastewater wetland” forests entrenched at intertidal zones representing one of the most diverse ecosystems in the world.²⁹ More than 90% of the world’s mangroves are located in developing countries, where human populations depend on their resources for livelihood.^{30,31} In coastal settings, mangroves receive domestic discharges and industrial effluents and they could be a reservoir of antibiotic-resistant genes (ARGs).^{32,33}

E. coli has acquired resistance to various antibiotics including third-generation antibiotics by the action of ESBL and so the degree of resistance will be elevated in upcoming years. As studies on ESBL resistance in mangrove ecosystems are relatively scarce, more research is required for identifying the occurrence of ESBL-harboring microbes. The objectives of this study were to determine the prevalence of ESBL gene in the *E. coli* isolates from the water and sediment samples of urbanized mangrove ecosystems in Kerala.

Materials and Methods

Description of the study area

Water and sediment samples were collected from three stations: Koduvallippuzha (station 1), Puthuvype (station 2), and Vallarpadam (station 3) based on their proximity to

anthropogenic inputs (Fig. 1). *Rhizophora mucronata* and *Avicennia marina* were the mangroves present in the study area. Station 1 receives hospital discharges, whereas stations 2 and 3 receive contaminants from industrial effluents and domestic sewages. Chinese dip-net operations and aquacultural practices are prevailing in all stations.

Sample collection

Water and sediment samples were collected from three stations monthly from December 2018 to March 2019. The water samples were collected in 1 L sterile plastic bottles (Tarson, India) 1 ft below the surface to get a better depiction of the water column. Sediment samples were collected using Van-Veen grab and packed in a Ziploc plastic bag. Collected samples were transported to the laboratory in an ice box and subjected to bacteriological examination within 4 hours of collection.

Isolation and biochemical identification of *E. coli*

Most probable number load of fecal coliform bacteria was determined by a three-tube dilution method using Lactose broth (Hi-Media, India) as the medium. For isolation of *E. coli*, one loopful from a positive lactose broth tube showing growth and gas production was streaked onto eosin methylene blue (Hi-Media, India) plates and incubated at 37°C for 24 hours. After incubation, typical *E. coli*-like colonies

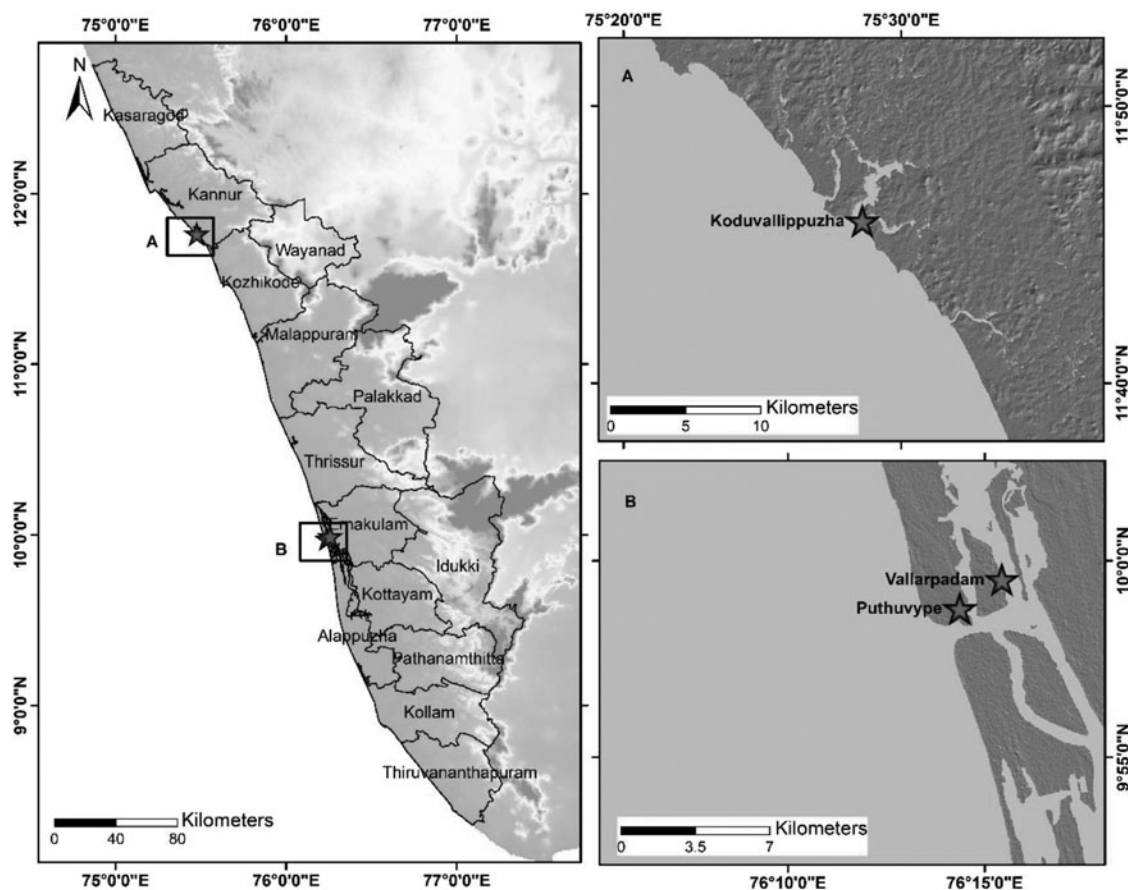


FIG. 1. Map representing the study area of Kerala and the sampling locations: Koduvallippuzha, Puthuvype, and Vallarpadam.

were selected, sub-cultured, and inoculated onto sterile nutrient agar slants (Hi-Media, India) vials for further characterization. For biochemical identification of *E. coli*, typical *E. coli*-like colonies were subjected to the IMViC test. The cultures giving ++-- reaction were confirmed as *E. coli*.

Molecular characterization of *E. coli*

Genomic DNA was extracted by the boiling method.³⁴ Plasmid DNA was extracted as per alkali lysis with the sodium dodecyl sulfate method: midiprep.³⁵ Rapid identification of *E. coli* isolates was done by multiplex PCR assay targeting conserved and species-specific gene: a 212 bp region of cyclic di-GMP regulator gene (*cdgR*).³⁶

Phylogenetic analysis of *E. coli*

E. coli phylogenetic groups (PGs) were determined by quadruplex PCR amplification of three genes (*arpA*, *chuA*, and *yjaA*) and a DNA fragment (TspE4.C2) for assigning *E. coli* isolate to one of the seven PGs (A, B1, B2, C, D, E, and F) according to Clermont et al.³⁷

Antibiotic susceptibility testing

Antibiotic resistance patterns of *E. coli* isolates were determined by the agar disk diffusion method according to Bauer et al.³⁸ The antibiotics used were ampicillin (AMP, 10 mcg), cefepime (FEP, 30 mcg), cefotaxime (CTX, 30 mcg), ceftaxidime (CAZ, 30 mcg), ciprofloxacin (CIP, 5 mcg), colistin (CST, 10 mcg), cotrimoxazole (COT, 25 mcg), doripenem (DOR, 10 mcg), ertapenem (ETP, 10 mcg), imipenem (IPM, 10 mcg), meropenem (MEM, 10 mcg), nalidixic acid (NAL, 30 mcg), piperacillin (PIP, 100 mcg), streptomycin (STR, 10 mcg), tetracycline (TET, 30 mcg), and trimethoprim (TMP, 5 mcg).

The agar dilution method was performed for reconfirming CST resistance. Pure cultures of *E. coli* were enriched in nutrient broth at 37°C for 6–8 hours and swabbed over the surface of dried sterile Mueller–Hinton agar plates. After 15–30 minutes of preincubation, antibiotic disks were aseptically placed over the seeded Mueller–Hinton agar plates, sufficiently separated from each other to avoid overlapping of the inhibition zone followed by overnight incubation at 37°C. After incubation, the diameter of the inhibition zone was measured and the isolates were classified as sensitive, intermediate, and resistant, according to the Clinical Laboratory Standards Institute (CLSI) guidelines.

Multiple antibiotic resistance indexing of *E. coli*

The multiple antibiotic resistance (MAR) index of an individual isolate was calculated by dividing the number of antibiotics to which the isolate was resistant by the total number of antibiotics to which the isolate was exposed.³⁹ The isolates with MAR index >0.2 were considered to have originated from a high-risk source of contamination, where antibiotics are often used in humans, livestock, and aquaculture.

PCR detection of extended-spectrum β -lactamases gene CTX-M-group 1

PCR targeting CTX-M-group 1 gene (688 bp) was carried out following Dallenne et al.⁴⁰ PCR products were analyzed

by electrophoresis on a 1.5% agarose gel, stained with ethidium bromide, and visualized by a gel documentation System (BioRad Gel Doc™ EZ Imager).

Statistical analysis

Pearson's chi-squared test was used to test the prevalence of antibiotic resistance and CTX-M-group 1 gene among sampling stations. Statistical analysis of the results was carried out using IBMSPSS version 22 (IBM Corporation). Statistical analysis was set at a *p*-value of <0.05.

Results

A total of 51 and 68 *E. coli* strains were isolated from water (station 1, 8 isolates; station 2, 16 isolates; station 3, 27 isolates) and sediment samples (station 1, 28 isolates; station 2, 22 isolates; station 3, 18 isolates), respectively.

PG distribution of *E. coli* strains

A total of 51 *E. coli* strains from water samples were subjected to phylogenetic analysis. Quadruplex PCR analysis of *E. coli* isolates (*n*=51) revealed that 29.4% of the isolates belonged to PG A, followed by PGs D (23.5%), B1 (15.6%), F (9.8%), and B2 (3.9%). Station 1 showed the highest percentage of incidence for PG A (37.5%) and PG F (25%). PGs B1 and D were predominant in station 2 (25% and 37.5% respectively). The incidence of PG B2 was shown exclusively by station 3 (7.40%). In sediment samples, 27.9% of the *E. coli* isolates (*n*=68) belonged to PG A, followed by PG B1 (26.4%), PG D (16.1%), PG E (4.4%), PG B2 (4.4%), and PG F (1.4%) (Fig. 2A–C).

Antibiotic resistance of *E. coli* isolates

All of the *E. coli* isolates from water (*n*=51) and sediment samples (*n*=68) were resistant to CTX and CST (100%) and were sensitive to IPM and MEM (Supplementary Table S1). In water samples, there was a significant difference in the prevalence of antibiotic resistance among three stations for antibiotics, namely AMP, CIP, DOR, ETP, PIP, STR, TET, and TMP (*p*<0.05). In sediment samples, there was a significant difference in the prevalence of antibiotic resistance among three stations for FEP, CAZ, COT, ETP, NAL, PIP, and CIP (*p*<0.05).

MAR index of *E. coli* isolates

MAR index was analyzed for *E. coli* strains isolated from water and sediment samples. MAR index values varied from 0.25 to 0.75 in station 1, 0.25 to 0.5 in station 2, and 0.25 to 0.68 in station 3 (Supplementary Tables S2–S7). In water samples, there was very high diversity among the antibiotic resistance patterns among *E. coli* isolates from station 3, indicating the possible origin of these strains from different sources. In sediment samples, MAR index values varied from 0.25 to 0.62 in station 1, 0.25 to 0.56 in station 2, and 0.31 to 0.68 in station 3.

Prevalence of CTX-M-group 1 gene among *E. coli* isolates

The prevalence of CTX-M-group 1 gene among the *E. coli* strains from three stations was analyzed (Supplementary

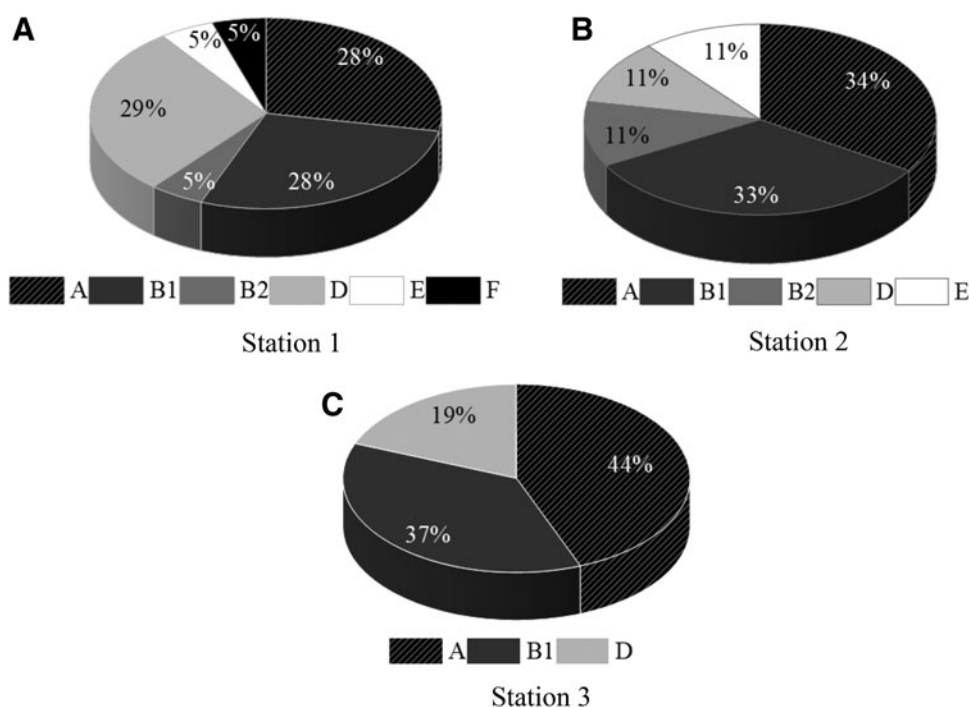


FIG. 2. (A–C) Phylogenetic distribution among *Escherichia coli* isolates from sediments of different mangrove stations.

Table S8). In water samples, a total of 30 *E. coli* strains harbored CTX-M-group 1 gene, where the prevalence was higher in station 3 (70.3%), followed by station 2 (50.0%) and station 1 (37.5%) ($p < 0.05$). In sediment samples, a total of 29 *E. coli* isolates harbored CTX-M-group 1 gene. Prevalence of CTX-M-group 1 was higher in station 1 (46.4%), followed by station 3 (44.4%) and station 2 (40.9%) ($p \geq 0.05$).

Discussion

Mangrove ecosystems are frequently encountered by human interventions and are prone to harbor ARB as they receive hospital, domestic, and aquacultural discharges. Several studies have focused on *E. coli* isolates from various environments to achieve a greater understanding of their origin, dissemination, and persistence of multidrug resistance.^{20,23,25,27,41} To our knowledge, this is the first study carried out in the water and sediment samples of three urbanized mangrove ecosystems of Kerala with respect to ESBL gene CTX-M-group 1.

In our study, a total of 51 and 68 *E. coli* strains were isolated from the water and sediment samples. *E. coli* is a member of the fecal coliform group of bacteria causing a wide variety of diseases, and their ability to survive and grow in the environment is likely due to their versatility in energy acquisition.^{42,43} The presence of *E. coli* in the mangrove ecosystem is a strong indicator of sewage or animal waste contamination.⁴⁴

Phylogenicity has been reported to play a significant role in the virulence of pathogens.^{45,46} In this study, PGs A and D were prominent in water, whereas PGs A and B1 were prominent in sediment samples. The most encountered group of *E. coli* isolates was PG A (29.4% in water and 27.9% in sediment samples). It is interesting to note that the highest percentages of PG A *E. coli* isolates (51 isolates, 29.4%) were isolated from station 1 that receives hospital effluents.

The complex balance between the commensalism and virulence is dependent on the host status and the expression of virulence determinants by bacteria.⁴⁷ PGs A and B1 are the nonpathogenic groups that occasionally cause extra-intestinal infections, which are devoid of virulence determinants than either B2 or D strains.^{47,48} Whereas PG B2 and, to a lesser extent, PG D possess virulence determinants that are responsible for many extraintestinal infections.^{46,49}

E. coli isolates of PGs A and B1 were the most frequently encountered phlotypes in this study, which agree with previous studies conducted in different aquatic environments.^{20,50,51} PG B1 possesses high number of ARG and plays a major role in the dissemination of antibiotic resistance in the aquatic environment.²² PG B2 can be more associated with anthropogenic contamination.^{22,52}

Some studies reported the link between the phlotype to their virulence, antibiotic resistance, and ARG.^{53–56} The prevalence of CTX-M-group 1 gene from sediment samples (42.6%) was found to be comparatively less when compared with water samples (58.8%). In water samples, *E. coli* isolates of PG D ($n = 10$), PG A ($n = 4$), PG F ($n = 3$), and PG B1 ($n = 3$) harbored ESBL gene CTX-M-group 1 in this study. In sediment samples, *E. coli* isolates of PG A ($n = 5$), PG B1 ($n = 5$), PG D ($n = 4$), and PG B2 ($n = 2$) harbored CTX-M-group 1 gene.

Previous studies have reported lower antibiotic resistance in PGs B2 and D.^{54,55} Various authors have reported multidrug resistance in PG D of clinical origin rather than environmental.^{53,56} Several studies have reported the prevalence and frequent co-occurrence of PGs A and B1 *E. coli* in an environment subjected to anthropogenic contamination.^{46,54,56,57} The variation in PGs in different study stations could be attributed to the different origins of *E. coli*.

In this study, all *E. coli* strains ($n = 119$) from the three stations were resistant to CTX and CST (100%) and were sensitive to IPM and MEM. The high frequency of antibiotic resistance among *E. coli* isolates in this study was in

agreement with previous studies conducted in mangrove ecosystems.^{22,28,58} Various studies have reported low incidence of CST and CTX in different environments.^{59–61} An earlier study reported that all the isolates from the mangrove swamps showed an MAR value of >0.2 and proved high-risk sources of contamination of the environment.²³

All *E. coli* isolates were carriers of multidrug resistance in our study (Supplementary Tables S3–S8). *E. coli* strains from three different mangroves from Rio de Janeiro showed multiresistance to several antimicrobial agents with an MAR index ranging from 0.12 to 0.31, indicating a comparatively low MAR value compared with the present study.⁶² A high frequency of phylotypes B2 and D in the tropical estuary of Cochin indicated that *E. coli* harboring multidrug resistant genes could pose a threat to those who rely on these aquatic ecosystems for subsistence.²⁴

Untreated sewage disposal from the villages together with active fish and shrimp farming contributed to the high prevalence of multidrug resistant *E. coli* in the mangrove ecosystem of Malaysia.⁵⁹ The prevalence of ARG has elevated mainly due to the abuse of antibiotics in human and veterinary medicine, as well as anthropic activities including sewage disposal.^{43,63} The phylotype and antibiotic susceptibility of *E. coli* vary significantly among the mangrove stations studied. β -Lactam antibiotics are widely recognized antibacterial agents because of their strong bactericidal activity and low toxicity.^{4,9}

The ESBL pandemic has significantly contributed to the rapid global increase in the rate of antibiotic resistance in *E. coli*. CTX-M-group 1 gene was present across all the phylotypes and can intensify therapeutic confrontation. High prevalence of phylotypes B2 and D can cause potential risks for humans as intense aquaculture practices are prevailing in these mangrove ecosystems.

Conclusion

Our study demonstrates the presence of *E. coli* with the ESBL gene CTX-M-group 1 in the mangrove environment and underlines the effects of anthropogenic disruptions on these highly productive and dynamic ecosystems. Aquaculture systems, as well as human and animal health, can be significantly impacted by pathogenic and multidrug-resistant *E. coli* isolates. Use and abuse of broad-spectrum antibiotics in human health care and veterinary medicine for livestock management are two key contributors to the emergence of antibiotic resistance.

For a precise understanding of antibiotic resistance in mangrove ecosystems, additional research focusing on ESBL resistance among bacterial isolates is required. For the purpose of keeping track of evolving antibiotic resistance patterns and ARB, the mangrove ecosystem's assessment of antibiotic resistance is of utmost importance.

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Authors' Contributions

G.S. contributed to conceptualization, investigation, formal analysis, data curation, and writing—original draft. D.P.S. was involved in conceptualization, formal analysis,

data curation, and writing—review and editing. E.A.P. carried out conceptualization and writing—review and editing. A.A.M.H. was in charge of conceptualization and writing—review, editing, and supervision.

Disclosure Statement

No competing financial interests exist.

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Supplementary Material

Supplementary Table S1
Supplementary Table S2
Supplementary Table S3
Supplementary Table S4
Supplementary Table S5
Supplementary Table S6
Supplementary Table S7
Supplementary Table S8

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